

(c) Reversible reactions and equilibria	
Students should:	
3.17	know that some reactions are reversible and this is indicated by the symbol $\rightleftharpoons$ in equations
3.18	describe reversible reactions such as the dehydration of hydrated copper(II) sulfate and the effect of heat on ammonium chloride
3.19C know that a reversible reaction can reach dynamic equilibrium in a sealed container	
3.20C know that the characteristics of a reaction at dynamic equilibrium are: • the forward and reverse reactions occur at the same rate • the concentrations of reactants and products remain constant.	
3.21C understand why a catalyst does not affect the position of equilibrium in a reversible reaction	
3.22C know the effect of changing either temperature or pressure on the position of equilibrium in a reversible reaction: • an increase (or decrease) in temperature shifts the position of equilibrium in the direction of the endothermic (or exothermic) reaction • an increase (or decrease) in pressure shifts the position of equilibrium in the direction that produces fewer (or more) moles of gas <i>References to Le Chatelier's principle are not required</i>	

4 Organic chemistry

The following sub-topics are covered in this section.

- (a) Introduction
- (b) Crude oil
- (c) Alkanes
- (d) Alkenes
- (e) Alcohols
- (f) Carboxylic acids
- (g) Esters
- (h) Synthetic polymers